

Name: _____

Exam Style Questions

3D Trigonometry



Equipment needed: Pen, Calculator, Ruler, Pencil

Guidance

1. Read each question carefully before you begin answering it.
2. Check your answers seem right.
3. Always show your workings

Video Tutorial

www.corbettmaths.com/contents

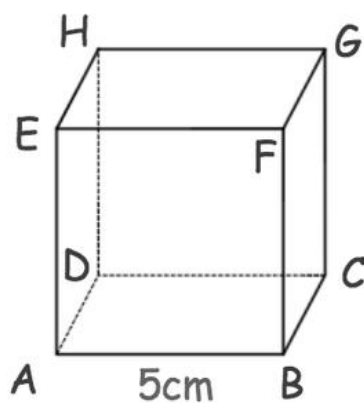
Video 332



Answers and Video Solutions



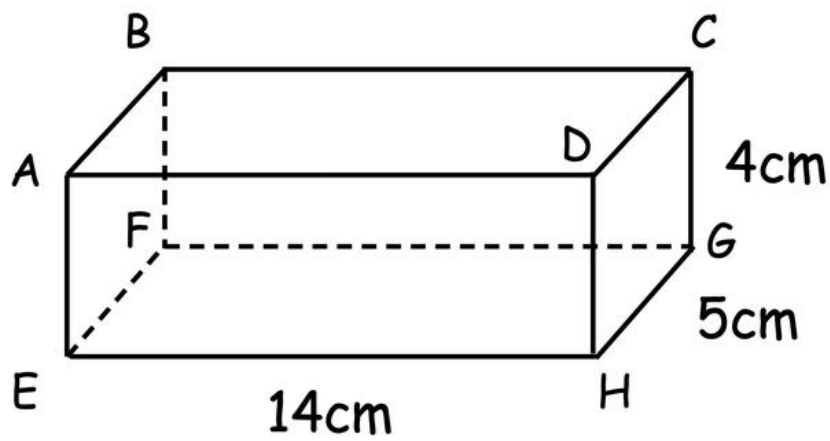
1. Shown is a cube with side length 5cm.



Calculate angle CAG.

.....°
(4)

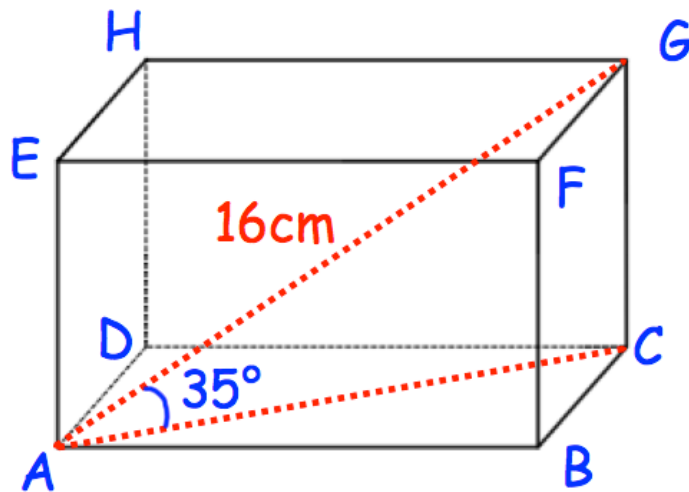
2. Shown below is a cuboid.



Calculate the size of angle CEG

.....°
(4)

3. Shown below is a cuboid.



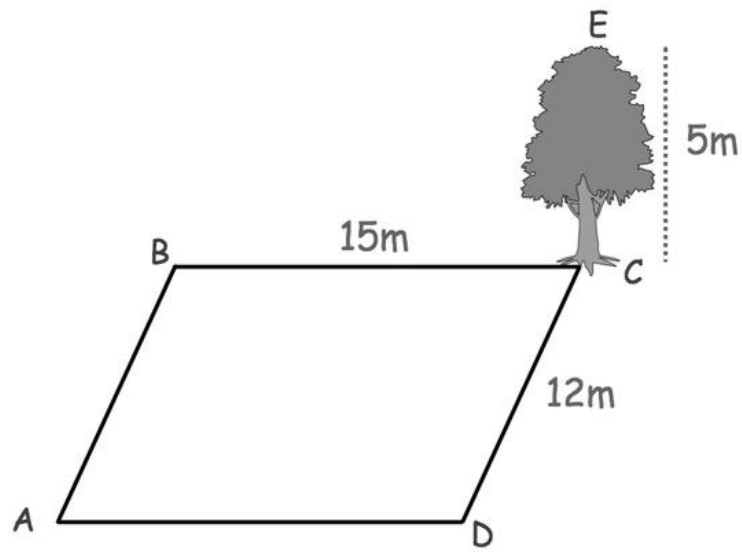
Length AG = 16cm

Angle CAG is 35°

Work out the length of EG.

.....cm
(3)

4. A tree is located in the corner of a rectangular field.

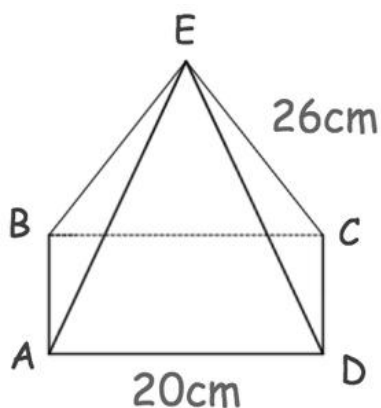


The field is 15 metres long and 12 metres wide.
The tree is 5 metres tall.

Calculate angle CAE.

.....°
(4)

5. Shown below is a square based pyramid.
The apex E is directly over the centre of the base.



$$AD = 20\text{cm}$$

$$CE = 26\text{cm}$$

- (a) Work out the length of AC

.....cm
(2)

- (b) Calculate angle CAE

.....°
(2)

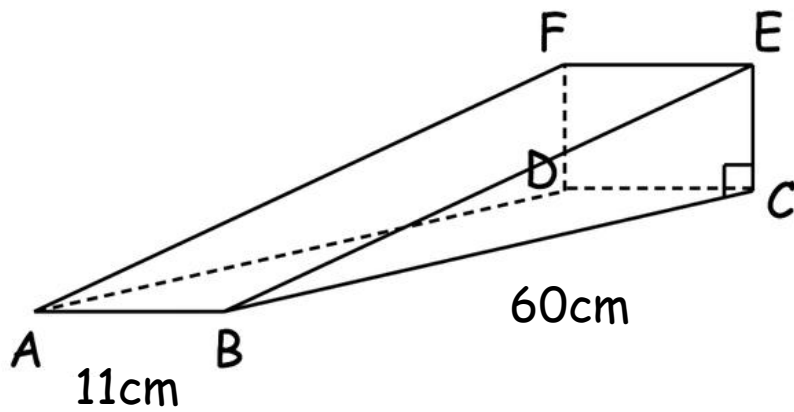
- (c) Work out the height of the pyramid

.....cm
(2)

- (d) Calculate the volume of the pyramid

.....cm³
(2)

6. A triangular prism is shown below.



Angle CBE = 9°

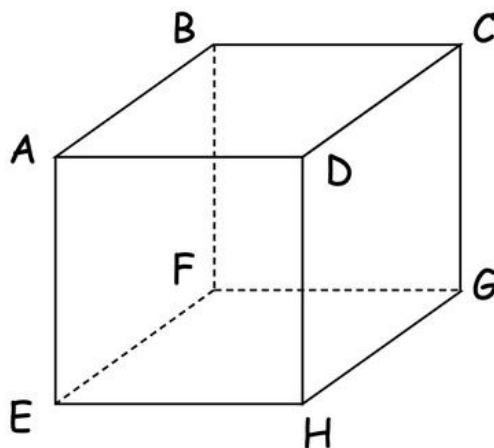
- (a) Calculate the length of BE.

.....cm
(2)

- (b) Calculate the length of BF.

.....cm
(2)

7. Shown below is a cuboid.



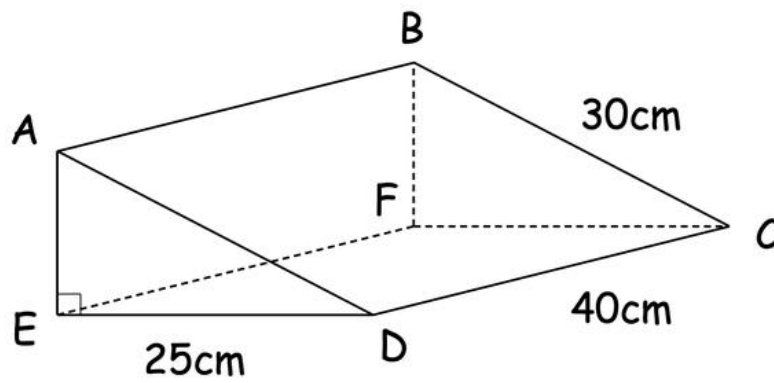
$$AE = 7\text{cm}$$

$$AG = 11.5\text{cm}$$

Work out the size of the angle between AG and plane EFGH.

.....°
(2)

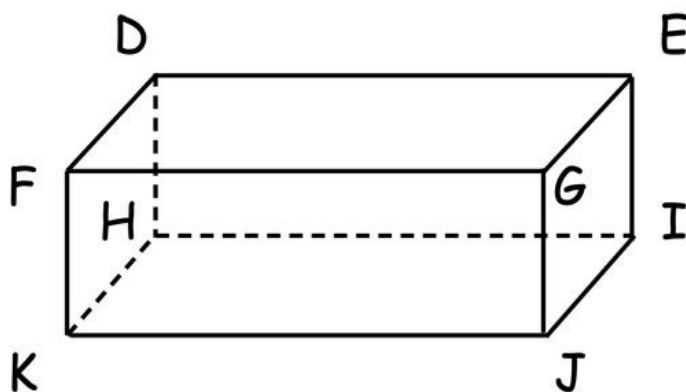
8. Here is a triangular prism.



Which angle is larger, angle ADE or angle ACE?
You must show your working.

(4)

9. Shown below is a cuboid.



$$KJ = 24\text{cm}$$

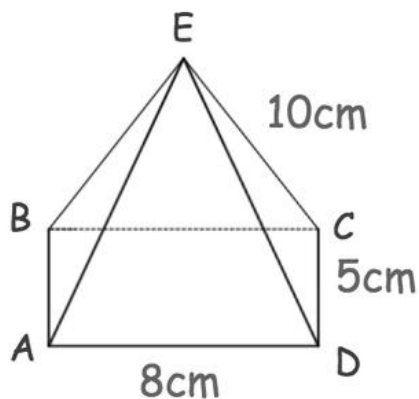
$$KG = 25\text{cm}$$

$$\text{Angle } HIK = 19.5^\circ$$

Calculate the size of angle KIF

.....°
(4)

10. Shown below is a rectangular based pyramid.
The apex E is directly over the centre of the base.



AD = 8cm
CD = 5cm
CE = 10cm

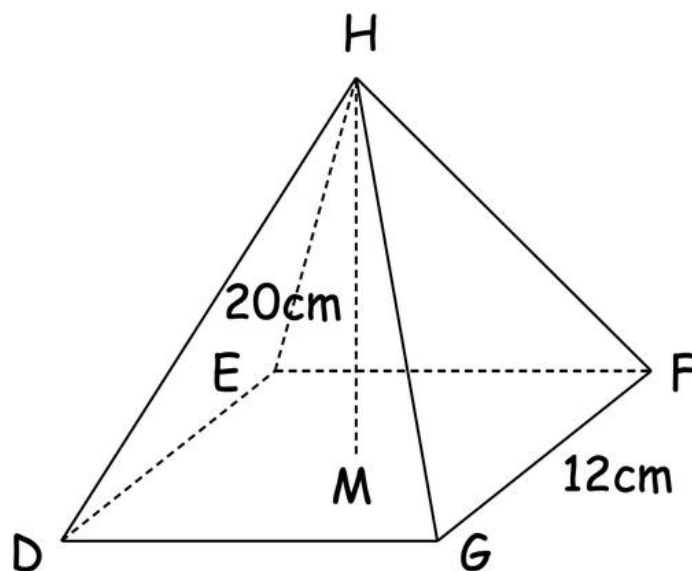
- (a) Calculate the height of the pyramid

.....cm
(4)

- (b) Calculate angle between the face ABE and the base ABCD

.....°
(3)

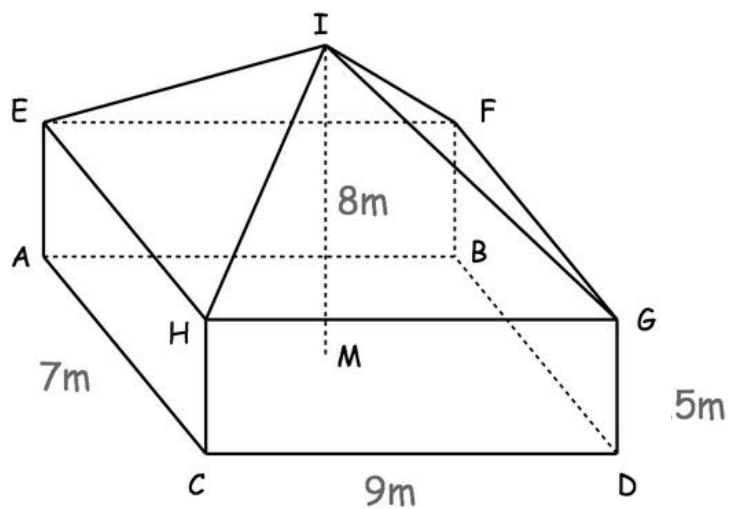
11. Shown below is a square based pyramid.
 The side length of the square is 12cm.
 The perpendicular height of the pyramid is 20cm.



Work out the size of angle HFD .

.....^o
(4)

12. The diagram shows a cuboid and a pyramid.
 The apex of the pyramid, I, is directly above the centre, M, of ABDC.



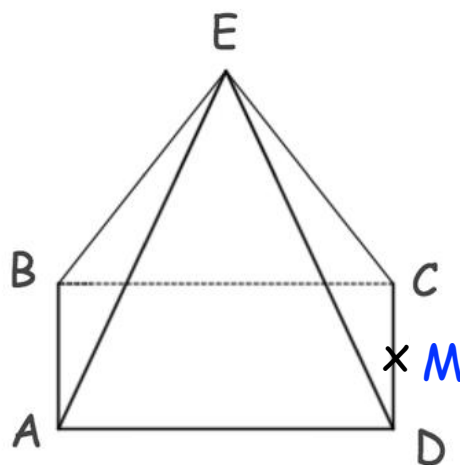
- (a) Calculate the angle between the line DI and the plane ABDC

.....°
(3)

- (b) Calculate the angle between planes EHI and ACHE

.....°
(2)

13. Shown below is a rectangular-based pyramid.
The apex E is directly over the base of the pyramid.



$$AD = 9\text{cm}$$

$$CD = 7\text{cm}$$

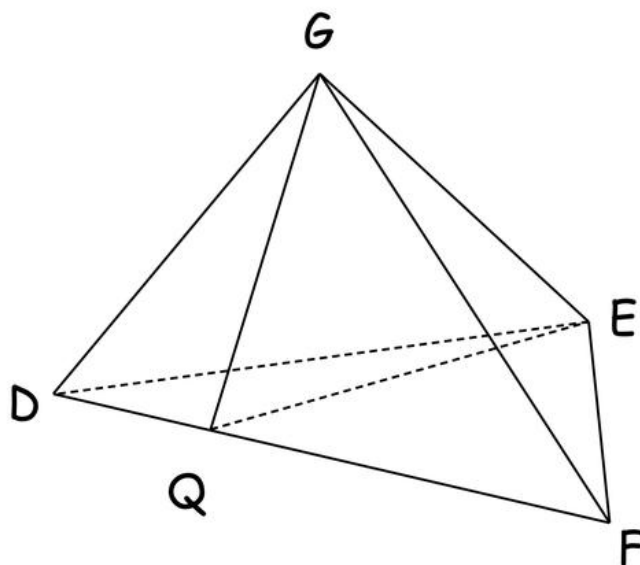
$$CE = 12\text{cm}$$

M is the midpoint of the line CD.

Work out the size of angle AME

.....°
(7)

14. DEFG is a triangular based pyramid.



Q is a point on DF.

$$GE = 6\text{cm}$$

$$\text{Angle } GEQ = 67^\circ$$

$$\text{Angle } GQE = 72^\circ$$

Work out the length of GQ.

.....cm
(3)